Conditional time-to-attack estimation accuracy

Competing-risk LSTM versus TCN | 3,375 true-event windows per model | Prediction horizon: 1–15 minutes

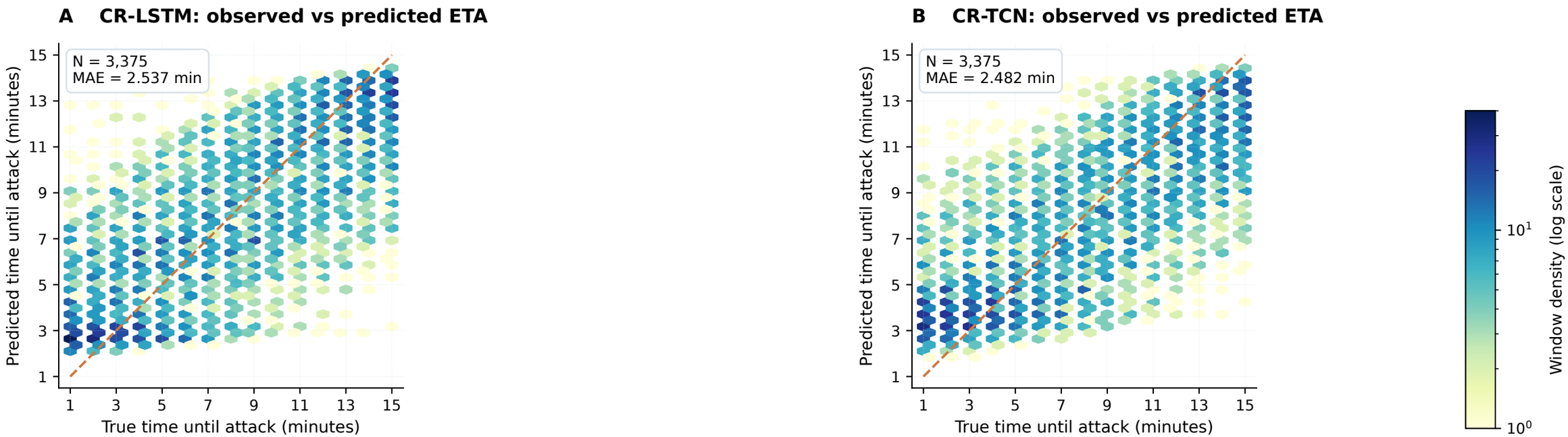

Figure 4. Conditional expected time-to-attack predictions from saved CR-LSTM and CR-TCN development-validation results.

Density panels use a common logarithmic count scale. The dashed diagonal represents perfect ETA prediction; the CDF shows absolute error.

Positive signed error means the model predicts more time remaining than actually exists; negative error means it predicts less.

Synthetic NAF v2 development validation. ETA is evaluated conditional on a true future attack. Windows overlap in time; confidence intervals and end-to-end onset accuracy are not implied.